

#1

ALTER TABLE passengers
ADD CONSTRAINT check_passenger_age
CHECK (age >= 10);

The screenshot shows a database IDE interface with the following components:

- Database Explorer:** Displays a tree view of database objects for 'task2@localhost', including tables like 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_flight', 'flights', 'passengers', 'security_check', and 'sequences'.
- SQL Editor:** Contains the following SQL script:

```
5 SET age = DATE_PART('year', AGE(CURRENT_DATE, date_of_birth));
6
7 SELECT passenger_id, first_name, last_name, date_of_birth, age
8 FROM passengers;
9
10 ALTER TABLE passengers
11 ADD CONSTRAINT check_passenger_age
12 CHECK ( age >= 10 );
13
14 SELECT conname AS constraint_name,
15        pg_get_constraintdef(c.oid) AS definition
16 FROM pg_constraint c
17 JOIN pg_class t 1:n<->1: ON c.conrelid = t.oid
18 WHERE t.relname = 'passengers';
19
```
- Output/Result Panel:** Shows the results of the last query (Result 14) in a table with 2 rows:

constraint_name	definition
passengers_pkey	PRIMARY KEY (passenger_id)
check_passenger_age	CHECK ((age >= 10))
- Services Panel:** Lists database services including 'lab2@localhost', 'console_3', 'task2@localhost', 'console_1', and 'passengers'.
- Bottom Bar:** Displays system information such as temperature (10°C), time (17:38), and date (10.10.2025).

#2

ALTER TABLE booking

ADD CONSTRAINT chk_ticket_price

CHECK (ticket_price >= 0 AND ticket_price <= 50000);

The screenshot shows a database management interface with a SQL editor and a results pane. The SQL editor contains the following code:

```
DELETE FROM booking
WHERE ticket_price < 0 OR ticket_price > 50000;

UPDATE booking
SET ticket_price = 50000
WHERE ticket_price > 50000;

UPDATE booking
SET ticket_price = 0
WHERE ticket_price < 0;

ALTER TABLE booking
ADD CONSTRAINT chk_ticket_price
CHECK (ticket_price >= 0 AND ticket_price <= 50000);
```

The results pane displays the definition of the constraints:

constraint_name	definition
1 booking_pkey	PRIMARY KEY (booking_id)
2 fk_booking_passenger	FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id)
3 fk_booking_flight	FOREIGN KEY (flight_id) REFERENCES flights(flight_id)
4 chk_ticket_price	CHECK (((ticket_price >= (0)::numeric) AND (ticket_price <= (50000)::numeric)))

The interface also shows a Database Explorer on the left with a tree view of database objects, and a Services pane at the bottom left showing transaction details.

#3

```
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK ( weight_in_kg BETWEEN 1 AND 23 );
```

The screenshot shows a database management tool interface. The top panel displays the 'Database Explorer' on the left, showing a tree view of database objects including 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_flight', 'flights', 'passengers', 'security_check', 'sequences', and 'Server Objects'. The main editor shows a SQL script with the following content:

```
55 WHERE weight_in_kg > 23;
56
57 UPDATE baggage
58 SET weight_in_kg = 1
59 WHERE weight_in_kg < 1;
60
61
62 ALTER TABLE baggage
63 ADD CONSTRAINT check_baggage_weight
64 CHECK ( weight_in_kg BETWEEN 1 AND 23 );
65
66 SELECT conname AS constraint_name,
67 pg_get_constraintdef(c.oid) AS definition
68 FROM pg_constraint c
69 JOIN pg_class t ON c.conrelid = t.oid
```

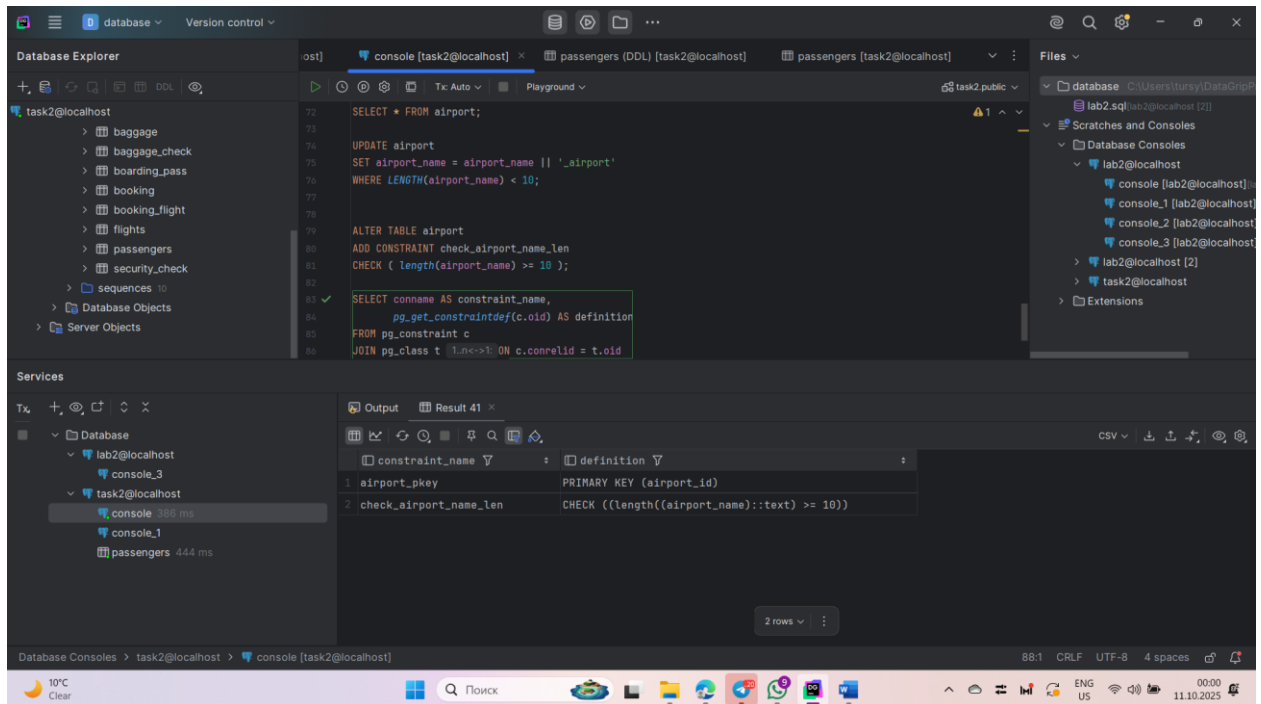
The bottom panel shows the 'Output' window with the results of the SQL execution. The results are displayed in a table with two columns: 'constraint_name' and 'definition'. The table contains three rows:

constraint_name	definition
baggage_pkey	PRIMARY KEY (baggage_id)
fk_baggage_booking	FOREIGN KEY (booking_id) REFERENCES booking(booking_id)
check_baggage_weight	CHECK (((weight_in_kg >= (1)::numeric) AND (weight_in_kg <= (23)::numeric))))

The bottom status bar shows the database console output, including the command 'task2@localhost' and the console name 'console [task2@localhost]'. The system tray at the bottom indicates the date and time as 10.10.2025, 23:53.

#4

```
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_len
CHECK ( length(airport_name) >= 10 );
```

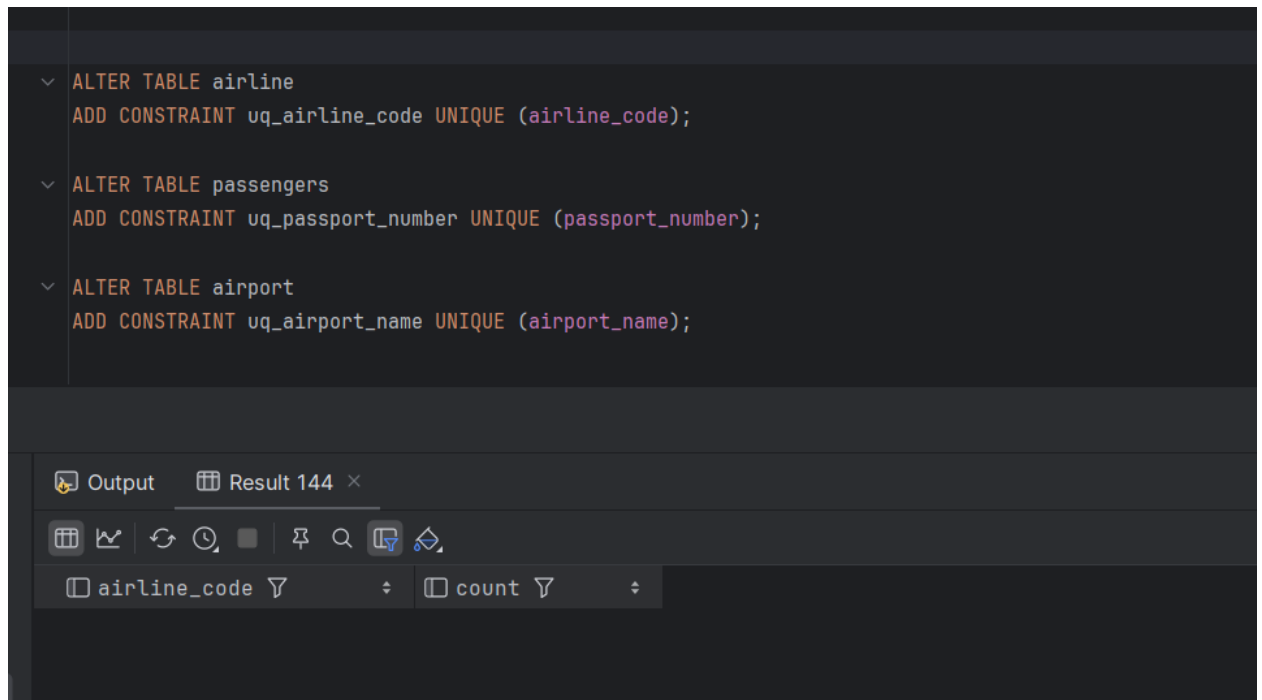


#5

```
ALTER TABLE airline
ADD CONSTRAINT uq_airline_code UNIQUE (airline_code);
```

```
ALTER TABLE passengers  
ADD CONSTRAINT uq_passport_number UNIQUE (passport_number);
```

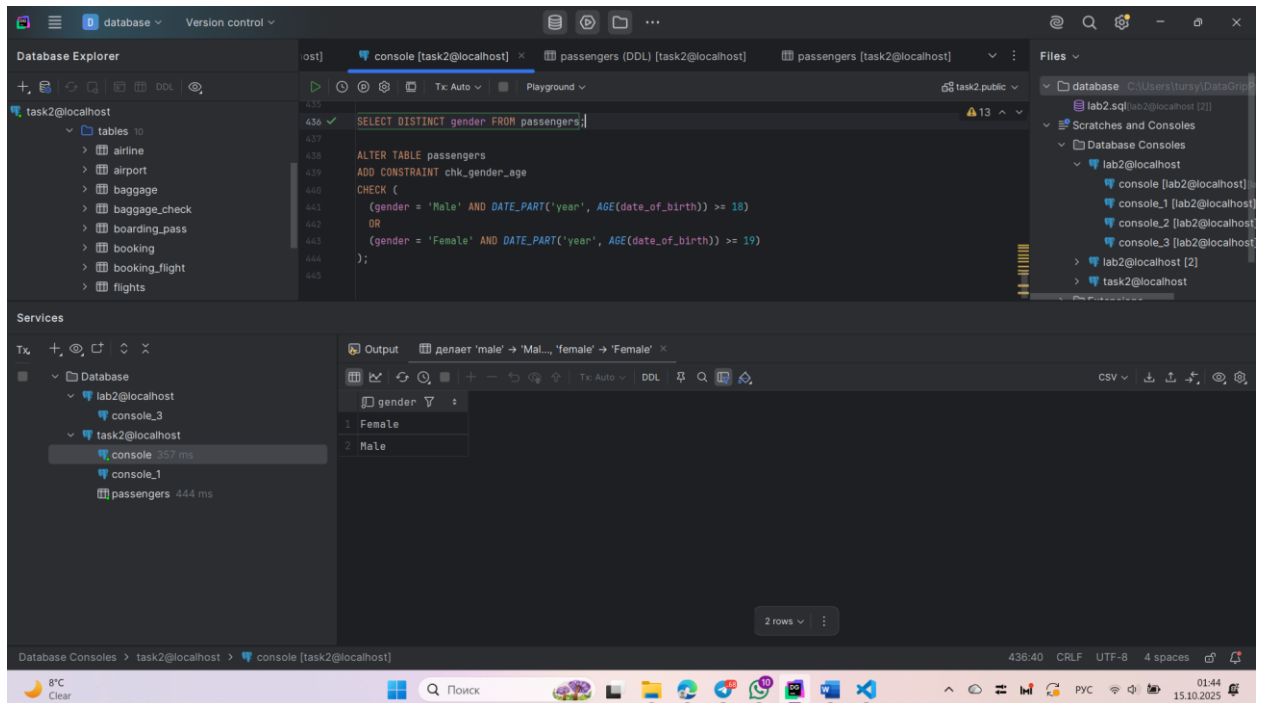
```
ALTER TABLE airport  
ADD CONSTRAINT uq_airport_name UNIQUE (airport_name);
```



#6

```
ALTER TABLE passengers  
ADD CONSTRAINT chk_gender_age  
CHECK (
```

```
(gender = 'Male' AND DATE_PART('year', AGE(date_of_birth)) >= 18)
OR
(gender = 'Female' AND DATE_PART('year', AGE(date_of_birth)) >= 19)
);
```



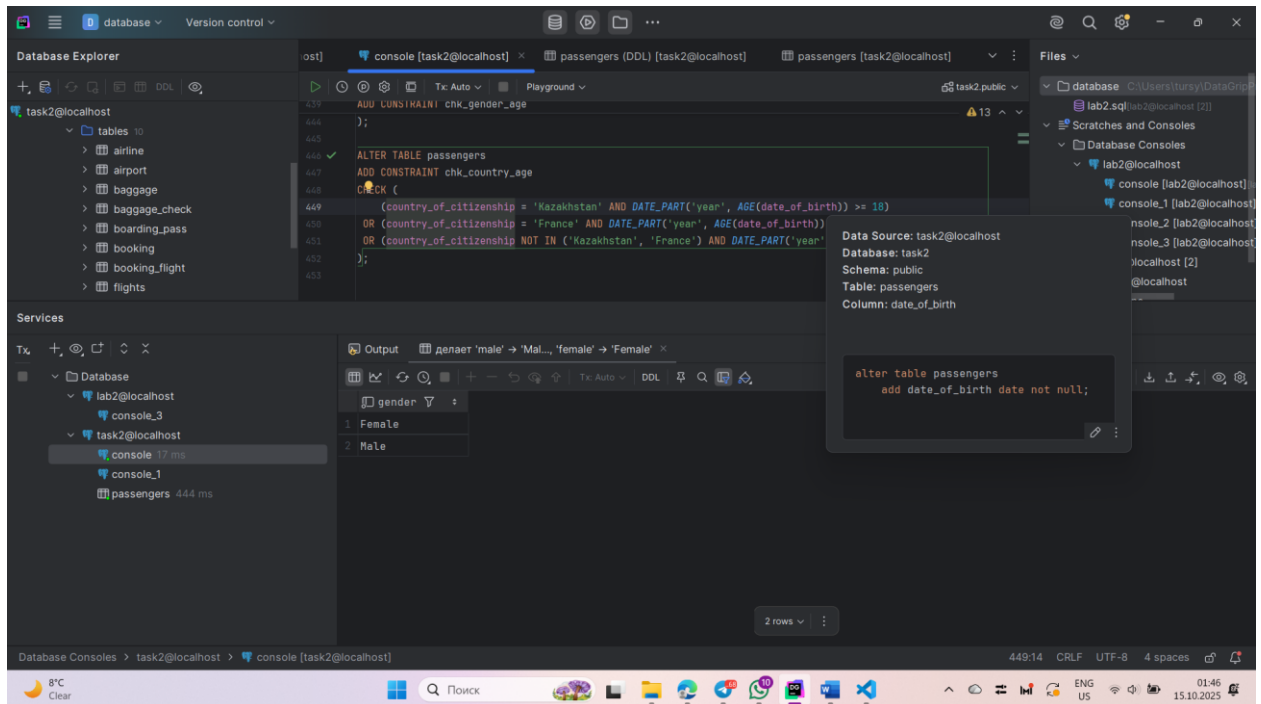
#7

```
ALTER TABLE passengers
ADD CONSTRAINT chk_country_age
CHECK (
```

```

(country_of_citizenship = 'Kazakhstan' AND DATE_PART('year', AGE(date_of_birth)) >= 18)
OR (country_of_citizenship = 'France' AND DATE_PART('year', AGE(date_of_birth)) >= 17)
OR (country_of_citizenship NOT IN ('Kazakhstan', 'France') AND DATE_PART('year',
AGE(date_of_birth)) >= 19)
);

```



#8

```

ALTER TABLE booking
ADD COLUMN ticket_discount DECIMAL(5,2);

```

```
ALTER TABLE booking
ADD CONSTRAINT chk_ticket_discount
CHECK (
    (created_at >= '2024-01-01' AND ticket_discount = ticket_price * 0.05)
    OR (created_at < '2024-01-01' AND ticket_discount = ticket_price * 0.10)
);
```

